Inference-Time Scientific Reasoning for Vision-Language Models on mmJEE-Eval: Code Sandboxes, Agentic Correction, Learned Verification and Retrieval, Tested on a Clean Exam Year

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***Abstract*—mmJEE-Eval shows that open vision-language models (VLMs) solve few JEE-Advanced problems and rarely fix their own errors. We wrap a *frozen* 4-bit Qwen3-VL-8B in a training-free framework: a code sandbox for program-of-thought solving, a cross-model Critic/Corrector loop, a learned verifier that selects the final answer, and retrieval of verified worked solutions. Our earlier version (v2) was developed with the 2025 exam as test set. Because 2025 was also used for piloting, and the model may have seen it, we froze an improved version (v3) on 2019–2024 data only and evaluated both versions on a *clean* test set: JEE-Advanced 2026 (204 questions, English and Hindi), which postdates the model. On 2026 the v3 framework reaches 58.8% (95% CI [50.0, 67.2]) vs. 44.3% for chain-of-thought, and beats self-consistency at the same generated-token budget by +6.4 points (CI [+1.5, +10.8], McNemar p=0.011). The frozen v2 framework reaches 55.4% and also beats matched self-consistency (+6.4). So the central claim holds on unseen data. v3 improves the solver (+6.8 points for PoT), but its overall advantage over v2 (+3.4) is not significant. On 2026 the new verifier selects no better than majority voting over the same pool, so the gain comes from a better candidate pool. A post-hoc final phase, with gates fixed in advance on 2019–2024, tested a fine-tuned neural verifier, length-robust and stacked selectors, training pools matched to the test-time solutions, method-card exemplars and an open-licensed knowledge base. None passed its gate or improved retrieval, so the v3 result stands; the remaining errors are mostly diagram questions whose right answer is rare or absent in the pool. Three last pre-registered checks show why. The gain over self-consistency comes from the candidate pool and is stable across years (+7.4 [+0.0, +15.3] points on 2025, +6.9 [+2.0, +11.8] on 2026), while the verifier's selection gain does not transfer, although its candidate AUC is higher on 2026. An answer-level verifier and a diagram-first transcript both fail their gates.**

I. Introduction

JEE-Advanced questions combine a diagram, multi-step physical or chemical reasoning, and exact answers. mmJEE-Eval [1] turns them into an image-only, bilingual benchmark and reports two weaknesses of open VLMs: low accuracy, and single-pass self-correction that turns a detected error into a correct answer only 1.1–5.2% of the time.

Our SOP proposed to attack both at inference time, without fine-tuning. The v2 system did so on the 2025 exam. Two threats remained. First, 2025 was used to pilot the system, so design choices may fit it. Second, Qwen3-VL may have seen the 2025 papers. This report addresses both:

- A **clean test year**. JEE-Advanced 2026 postdates the model. Its English questions are in the dataset; we built the Hindi twins ourselves from the official papers.
- A **pre-registered freeze**. Every v3 change was chosen on 2019–2024 data, by leave-one-year-out (LOYO) cross-validation or on the 2024 dev year, with inclusion rules fixed in advance. It was then frozen (`V3_FREEZE.md`) before any 2026 or 2025 run.
- **Both frameworks on 2026**. The frozen v2 system is re-run unchanged next to v3, with compute-matched self-consistency (SC) [2], cluster-bootstrap CIs and McNemar tests for every claim.
- **Analyses**: correction dynamics in the paper's terms, a learned correction gate, the verifier's distribution shift, a bridge to the paper's Qwen2.5-VL-7B, and memorisation probes.
- A **post-hoc final phase** (Sec. VII) that attacks the two weak points of v3, selection and retrieval, under a second pre-registration (`V4_PREREG.md`).

II. Related work

Tool-augmented reasoning. Program-of-thought [3] and PAL [4] move arithmetic into an interpreter. **Sampling and voting.** Self-consistency [2] votes over sampled chains and is our compute-matched baseline. **Self-correction.** Self-Refine [5] and CRITIC [6] iterate critique and revision, but intrinsic self-correction often fails without external signal [7]. We add code execution, step localisation and a cross-model critic. **Verifiers.** Outcome and process verifiers [8], [9] rank sampled solutions; ours is a light gradient-boosted ranker trained on automatically labelled training-year pools. **Retrieval.** RAG [10] supplies context. We retrieve verified worked solutions with FAISS [11] and bge [12] embeddings.

III. Data and evaluation protocol

Splits. 2019–2024 (1,270 questions) are used for training and development; 2024 is the dev year. **Clean test:** 2026, 204 questions (102 English from the dataset plus their 102 Hindi twins). We cropped the Hindi twins from the official Hindi

papers, matched them by paper, subject and number, took answers from the English key. Automatic checks match every twin, and 14 crops, including every size outlier, were checked by hand against their English twins. Their style differs from the dataset's images, so results are also reported per language. **Post-hoc set:** 2025, 190 questions. It was the v2 test set and was used to pilot v1/v2, so it is reported separately and labelled post-hoc.

Scoring and metrics. We use the paper's `extract_answer` and `is_answer_correct`, copied verbatim and checked by tests. Pass@1 is the mean over samples. Selected-answer rows report the accuracy of the selected answer. CIs use a cluster bootstrap over EN/HI twin pairs (2,000 resamples), and differences use the paired version. McNemar treats questions as independent, so its p is optimistic. **Compute matching:** SC uses the smallest number of CoT samples whose generated tokens reach the framework's (capped at the 20 samples drawn).

Leakage control. No test question or twin is in any training pool. KB entries close to any 2025 or 2026 question are removed (de-duplication against both years). The v3 freeze used no 2025 or 2026 data.

IV. Method

Shared core (v2). *A. Code sandbox:* the solver writes numbered steps and puts every calculation in a `python` block. The block runs in an isolated subprocess (AST whitelist, no I/O, resource limits), and its output is fed back before generation continues. If no `\boxed{}` answer appears, a plain-CoT fallback answers. *B. Correction:* a cross-model critic (InternVL3.5-8B [15]) names the first wrong step or says `NO_ERROR`. The corrector keeps the earlier steps and re-derives from the flagged one, for up to 3 rounds; it stops early when the critic accepts the solution or the answer is unchanged for two rounds. *C. Verifier:* a classifier over handcrafted and embedding features picks one answer from 16 candidates (8 originals + 8 corrected). *D. Retrieval:* a one-line concept tag retrieves up to two verified worked solutions from 2019–2024 as exemplars.

v3 changes (Table I): (i) *Adaptive budget.* A cut-off answer is continued once for 4,096 more tokens. This applies to every row, including the baselines, so comparisons stay fair. (ii) *Transcript block.* The solver prompt also contains the VLM's own English transcription of the question (Hindi is translated), to help it read. (iii) *RAG v3.* Hybrid tag + problem-text similarity, a KB extended with JEEBench 2016–2018, and τ tuned by LOYO. LOYO coverage is 11.7% vs. 8.4% for the v2 recipe at the same proxy precision. (iv) *Verifier v3.* An XGBoost pairwise ranker, selected by LOYO over 2019–2024. It is trained on pools that, like the test pools, contain corrected solutions, and it gets critic features (flagged, rounds, answer changed) and agreement over the whole pool and its halves. LOYO pick accuracy is 62.0% vs. 53.7% for majority voting (oracle 78.5%). (v) *Correction gate.* A chain keeps its corrected answer only if the verifier score rises by more than $\delta{=}-0.08$, with δ chosen out-of-year. This is reported as its own row; the framework's final answer is still the verifier's pick from the full pool.

TABLE I
DEV A/B TESTS ON 2024 (204 QUESTIONS) THAT DECIDED THE FREEZE. RULE FIXED IN ADVANCE: INCLUDE IFF $\Delta > 0$ AND THE CI LOWER BOUND > 0.

Arm	Base	Arm	Δ [95% CI]
Transcript block (PoT)	50.2	55.1	+4.9 [+1.8, +8.2]
Adaptive budget (PoT)	50.2	52.6	+2.3 [+1.0, +3.7]
Adaptive budget (CoT)	48.5	54.9	+6.4 [+2.5, +10.8]
RAG, v2 recipe	50.2	50.4	+0.2 [−2.3, +2.9]
RAG v3 (hybrid + JEEBench)	50.2	50.6	+0.3 [−2.1, +3.0]

Setup. Qwen3-VL-8B-Instruct [14] and InternVL3.5-8B (AWQ 4-bit) run with vLLM [13] on one RTX 3060, one model at a time. Sampling uses $T{=}0.7$, top-p 0.8 and top-k 20, the same for all rows. CoT takes 20 samples per question; PoT and correction take 8. Every call is cached in SQLite and seeds are deterministic, so the three lab shutdowns during the final batch resumed without changing any result.

V. Results on the clean 2026 test

A. Main result

Table II and Fig. 2 give the SOP's Table I for both frameworks. The v3 framework reaches **58.8%** vs. 44.3% for single-pass CoT (+14.6 points). At 80k generated tokens per question, matched SC uses 18 samples and reaches 52.5%. The framework is +6.4 points better (CI [+1.5, +10.8]): it answers 18 questions that SC misses, against 5 the other way. The frozen v2 framework reaches 55.4%, also +6.4 points above its matched SC (CI [+1.5, +11.8], $p{=}0.015$). *The main v2 claim, beating compute-matched self-consistency, therefore holds on an exam the model cannot have seen.* Fig. 3 shows that every framework row lies above the SC curve at its cost.

B. v3 vs. v2

Table III gives the paired tests. The v3 solver is clearly better: v3 PoT beats v2 PoT by +6.8 points (CI [+3.6, +10.0]). The adaptive budget alone lifts CoT by +3.0. The full v3 framework is +3.4 points above v2 (CI [−1.5, +8.8], $p{=}0.230$), which is *not* significant on 204 questions. On the post-hoc 2025 set the same comparison is +6.8 (CI [+1.1, +12.1]). The difference between the two years is largely the verifier (Sec. V-E).

C. Where the frameworks win and lose

Table IV shows that v3 improves on most groups, but the gaps the paper reports remain. For v3 on 2026, English reaches 63.7% and Hindi 53.9%; text-only questions reach 72.5% and diagram questions only 30.3%. Inference-time reasoning does not fix perception. Part of the Hindi gap on 2026 may come from our own crops.

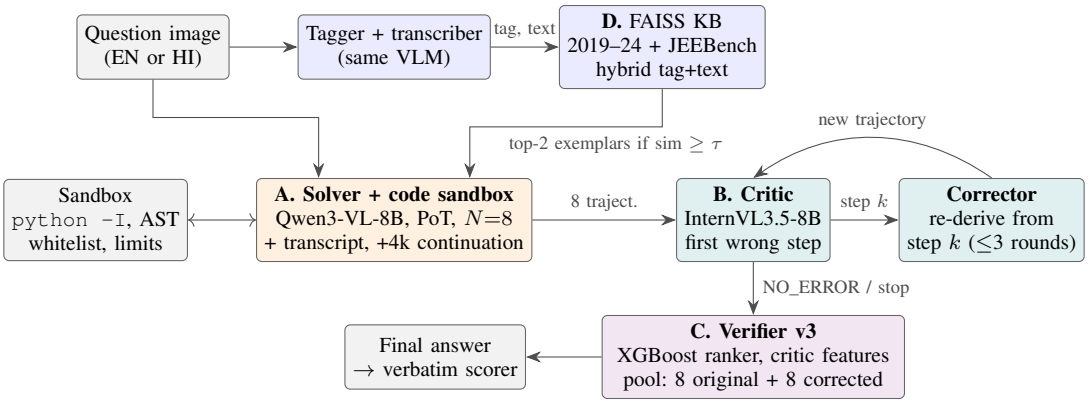


Fig. 1. The v3 framework around a frozen VLM. Only the verifier (C) is trained, on automatically labelled 2019–2024 pools. The v3 changes from v2 are the transcript, the continuation, hybrid retrieval with JEEBench, and the verifier v3 with its correction gate.

TABLE II

TABLE I ON THE CLEAN 2026 TEST SET (204 QUESTIONS). TOP: FROZEN V3 FRAMEWORK. BOTTOM: THE FROZEN V2 FRAMEWORK RE-RUN UNCHANGED. ACCURACY IN %; 95% CLUSTER-BOOTSTRAP CIS OVER EN/HI TWINS; TOKENS ARE GENERATED TOKENS.

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
v3 framework						
Baseline (single-pass CoT, adaptive budget)	44.3	29.5	37.4–51.0	–	4.5k	1.0
Self-consistency (matched compute) (n=18)	52.5	34.6	43.1–61.3	+8.2 [+4.5, +12.0]	80.4k	18.0
+ Code sandbox + RAG v3 + transcript (v3 PoT)	49.8	37.3	42.5–56.9	+5.6 [+2.0, +9.2]	6.5k	2.2
+ Learned verifier v3 (PoT pool only)	57.8	41.0	48.5–66.2	+13.6 [+8.4, +18.6]	51.8k	18.3
+ Correction (cross-model critic)	52.0	39.6	44.6–59.2	+7.7 [+4.3, +11.0]	10.0k	5.1
+ Correction (gated)	52.6	39.9	44.9–60.1	+8.3 [+4.7, +11.7]	10.0k	5.1
v3 framework (verifier v3)	**58.8**	44.9	50.0–67.2	+14.6 [+9.6, +19.5]	80.3k	41.7
v2 framework (re-run)						
Baseline (single-pass CoT)	41.2	26.5	34.4–47.9	–	2.9k	1.0
Self-consistency (matched compute) (n=20, needs 21)	49.0	28.2	40.2–57.4	+7.8 [+4.4, +11.3]	58.0k	20.0
+ Code sandbox	43.0	31.1	36.4–49.6	+1.8 [−1.1, +4.7]	4.0k	1.8
+ Agentic correction (cross-model critic)	46.8	36.2	40.1–53.4	+5.5 [+2.1, +9.0]	7.4k	4.7
+ Learned verifier	55.4	39.7	46.6–63.7	+14.2 [+9.0, +19.5]	59.2k	38.0
Full framework	**55.4**	39.7	46.6–63.7	+14.2 [+9.0, +19.3]	60.1k	38.8

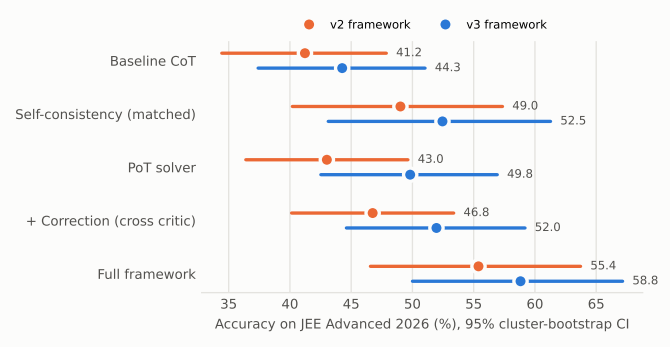


Fig. 2. 2026 clean test: matching rows of the v2 and v3 frameworks, with 95% cluster-bootstrap CIs.

TABLE III

PAIRED COMPARISONS (CLUSTER BOOTSTRAP; MCNEMAR COUNTS QUESTIONS ONLY a / ONLY b GETS RIGHT; IT IS MEANINGFUL FOR SINGLE-ANSWER ROWS ONLY, AS PASS@1 ROWS ARE ROUNDED PER QUESTION). †POST-HOC.

Comparison (a vs. b)	Δ [95% CI]	P(Δ≤0)	a/b	p
2026 v3 fw vs. v2 fw	+3.4 [−1.5, +8.8]	0.113	16/9	0.230
2026 v3 PoT vs. v2 PoT	+6.8 [+3.6, +10.0]	0.000	27/3	<0.001
2026 CoT adaptive budget vs. CoT	+3.0 [+1.7, +4.5]	0.000	6/3	0.508
2026 gated vs. ungated correction (v3)	+0.6 [−0.4, +1.5]	0.122	4/1	0.375
2026 verifier v3 pick vs. v3 PoT mean	+9.0 [+5.8, +12.4]	0.000	17/0	<0.001
2025† v3 fw vs. v2 fw	+6.8 [+1.1, +12.1]	0.012	21/8	0.024
2025† v3 PoT vs. v2 PoT	+8.1 [+4.8, +11.6]	0.000	30/7	<0.001

D. Correction loop and the correction gate

In the paper's terms (Table V), the cross-model critic flags 52.9% of wrong v3 solutions in round 1 (69.0% for v2). Round-1 correction fixes 13.2% of the detected errors (10.8% for v2), well above the paper's 1.1–5.2% single-pass rate. The critic also flags 19.9% of *correct* solutions. Because the v3 solver is stronger, fewer wrong solutions remain, and the correction loop gains less in absolute terms.

The gate (Table VI) does what it was trained for. On 2026 it cuts right→wrong flips from 33 to 9 while keeping most fixes (68→54 wrong→right). Per-chain accuracy goes from 52.0% to 52.6% (+0.6, CI [−0.4, +1.5]); the change is small and not significant. The flip pattern carries over from the training years to both test years.

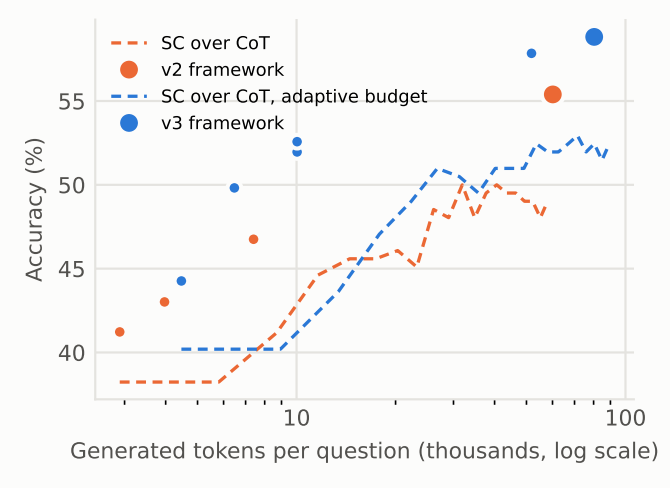


Fig. 3. 2026: accuracy vs. generated tokens. Dashed: SC over n CoT samples (majority vote). Dots: Table I rows; large dots: the full frameworks.

TABLE IV
FULL-FRAMEWORK ACCURACY (%) BY GROUP. 2025 IS POST-HOC.

Group	v2 '26	v3 '26	v2 '25	v3 '25
English	60.8	63.7	73.7	74.7
Hindi	50.0	53.9	56.8	69.5
Text only	69.6	72.5	72.5	81.9
Needs image	25.8	30.3	46.2	46.2
Physics	32.4	35.3	54.8	64.5
Chemistry	52.9	60.3	60.9	59.4
Mathematics	80.9	80.9	79.7	92.2
MCQ-Single	79.2	83.3	67.4	65.2
MCQ-Multiple	55.6	55.6	59.5	69.0
Numerical	39.7	44.9	69.0	77.4
Matching	58.3	62.5	55.6	72.2

TABLE V
CORRECTION LOOP, PAPER-STYLE METACOGNITION (%). EP: WRONG SOLUTIONS FLAGGED IN ROUND 1; EC: ROUND-1 FIXES AMONG DETECTED; FA: CORRECT SOLUTIONS FLAGGED. W→R / R→W: ANSWER FLIPS AFTER ALL ROUNDS.

Run	Chains	Acc.	EP	EC	FA	W→R	R→W
2026 pot_corr_cross	1632	43.0→46.8	67.5	9.7	23.6	76	15
2026 full	1632	44.7→48.8	69.0	10.8	25.2	83	16
2026 v3_full	1632	49.8→52.0	52.9	13.2	19.9	68	33
2025 v3_full	1520	56.2→57.9	40.3	13.1	23.3	46	21
2025 pot_corr_cross (v2)	1520	48.2→51.3	59.3	13.3	25.7	87	39
2025 full (v2)	1520	50.9→54.7	60.7	15.7	28.6	87	29

TABLE VI
CORRECTION GATE: PER-CHAIN ACCURACY (%) AND FLIPS, UNGATED → GATED.

Set	Chains	Orig.	Corr.	Gated	W→R	R→W
Train (LOYO, picks δ)	1936	44.4	47.6	**48.2**	95→88	32→14
2026 (clean)	1632	49.8	52.0	**52.6**	68→54	33→9
2025 (post-hoc)	1520	56.2	57.9	**58.5**	46→41	21→7

TABLE VII
PRIMARY VERIFIER ON ITS OWN 16-ANSWER TEST POOL: AUC AND ACCURACY OF THE PICKED ANSWER VS. POOL BASELINES (%).

Pool	Verifier	AUC	Pick	Maj.	Rand.	Oracle
2026 v3	xgb_ranker/handcrafted	0.902	**58.8**	59.3	50.9	79.4
2026 v2	logreg/all	0.913	**55.4**	53.9	46.7	76.5
2025 v3	xgb_ranker/handcrafted	0.872	**72.1**	65.8	57.1	87.4
2025 v2	logreg/all	0.836	**65.3**	65.8	52.8	85.3

TABLE VIII
SHIFT BETWEEN THE VERIFIER'S TRAINING SOLUTIONS AND THE TEST POOLS. CUT: `FINISH = LENGTH` ON THE FINAL CALL.

Solutions	Rows	Acc.	Cut	Tok.	Acc. cut	Acc. compl.
train 2019-2024 pot_train	10160	45.5	37.2	3.5k	28.9	55.3
2026 pot	1632	43.0	42.8	4.0k	27.0	55.0
2026 v3_pot	1632	49.8	40.3	6.5k	32.0	61.8
2025 pot	1520	48.2	36.7	3.5k	29.9	58.7
2025 v3_pot	1520	56.2	36.6	5.9k	41.7	64.7

E. Verifier: classifier vs. selector

As a classifier, verifier v3 transfers well: its AUC on the 2026 pool is 0.90. *As a selector, on the clean year, it adds nothing*: it picks 58.8% vs. 59.3% for majority voting over the same 16 answers (−0.5), with an oracle of 79.4%. On 2025 it beats majority by +6.3 points. So the v3 gain on 2026 comes from a better candidate pool (transcript, continuation, correction), not from selection. Picking among the 8 PoT samples alone already reaches 57.8%.

The shift stated before the runs is visible in Table VIII. The verifier was trained on plain PoT solutions of about 3.5k tokens (37.2% cut off). The 2026 v3 solutions are 6.5k long because of the transcript and continuation, and 40.3% are cut off. This is a plausible reason the selector gain does not carry over to the clean year.

VI. POST-HOC 2025, MODEL BRIDGE AND MEMORISATION

2025 (post-hoc). The v3 framework reaches 72.1% (Table IX) vs. 65.3% for the frozen v2 run, and beats matched SC by +13.7 points. Since 2025 shaped earlier design choices and may be in the model's training data, we do not base claims on it. The larger gaps on 2025 than on 2026 are consistent with some optimism on that year.

Bridge to the paper's model. With Qwen2.5-VL-7B (2025 is clean for this model), the framework does not help: 14.0% for CoT, 16.3% for matched SC and 15.3% for the framework (−1.1 vs. SC, CI [−7.4, +5.3]; Table X). At about 14% accuracy, most candidates are wrong, so agreement, critique and verification carry little signal. The framework needs a solver that is right often enough.

Memorisation probes. Asked for the answer only, Qwen3-VL scores 10.5–15.7% on 2019–2024, 13.2% on 2025 and 15.2% on the clean 2026 (Table XI). The clean year is not lower, so we see no memorisation signal. Pass@5 of v3 PoT on 2026 is 70.2%, well above any selector, so there is still headroom in the pool.

TABLE IX
V3 TABLE I ON 2025 (190 QUESTIONS, **POST-HOC**: 2025 WAS USED TO PILOT V1/V2).

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
Baseline (single-pass CoT, adaptive budget)	48.3	43.2	41.7–55.3	–	4.0k	1.0
Self-consistency (matched compute) (n=18)	58.4	52.4	50.0–66.8	+10.1 [+6.1, +14.1]	72.4k	18.0
+ Code sandbox + RAG v3 + transcript (v3 PoT)	56.2	59.7	49.5–63.0	+8.0 [+3.3, +13.1]	5.9k	2.3
+ Learned verifier v3 (PoT pool only)	68.9	72.6	61.1–76.8	+20.7 [+14.7, +27.3]	47.3k	19.5
+ Correction (cross-model critic)	57.9	62.5	51.1–64.5	+9.6 [+4.8, +15.0]	8.7k	5.0
+ Correction (gated)	58.5	63.7	51.7–65.1	+10.2 [+5.3, +15.6]	8.7k	5.0
v3 framework (verifier v3)	**72.1**	77.4	64.7–80.5	+23.8 [+17.4, +30.7]	69.9k	41.1

TABLE X
BRIDGE: THE FRAMEWORK WITH THE PAPER'S QWEN2.5-VL-7B ON 2025 (CLEAN FOR THAT MODEL; VERIFIER TRANSFERRED FROM QWEN3).

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
Baseline (single-pass CoT)	14.0	6.6	11.0–17.3	–	0.7k	1.0
Self-consistency (matched compute) (n=20, needs 33)	16.3	10.7	10.0–23.2	+2.3 [−1.9, +6.5]	14.1k	20.0
+ Code sandbox	13.9	5.7	10.2–18.0	−0.1 [−2.9, +2.9]	1.0k	3.6
+ Correction (same-model critic)	13.9	6.1	10.2–17.9	−0.1 [−2.9, +2.9]	1.5k	6.7
+ Correction (cross-model critic)	13.7	8.0	10.3–17.3	−0.3 [−2.9, +2.5]	2.8k	11.6
+ Learned verifier (transfer)	15.3	8.3	10.0–21.1	+1.3 [−3.3, +6.0]	22.7k	92.9

TABLE XI
MEMORISATION PROBE: ANSWER-ONLY ACCURACY (%, 64 TOKENS) BY EXAM YEAR. 2026 IS AFTER BOTH MODELS' RELEASE.

Model	'19	'20	'21	'22	'23	'24	'25	'26
Qwen3-VL-8B	12.5	14.6	11.8	10.5	14.5	15.7	13.2	15.2
Qwen2.5-VL-7B	12.5	12.3	14.5	17.1	13.5	14.2	12.1	21.1

VII. POST-HOC EXTENSIONS (PHASE 4)

After the v3 result we tried to fix its two weak points, selection and retrieval. Both test years had been seen by then, so *every Phase-4 number on 2025 or 2026 is post-hoc*. All choices used 2019–2024 only, and a second pre-registration (`V4_PREREG.md`), written before any Phase-4 scoring, fixed the rules: a new selector is scored on the test years only if its LOYO accuracy beats the v3 ranker's on every pool kind. The v3 result above stays the headline.

Diagnosis. Length shift is not the main cause of the 2026 selector failure. v3 solutions are longer than the training pools in *both* years (7.3k and 8.3k vs. 4.0k tokens), with small standardised shifts of log length (0.15 and 0.28). Yet the verifier beats majority on 2025 (+6.3 [+1.6, +11.1]) and not on 2026 (−0.5 [−2.9, +2.5]). On 2026 it loses mainly on pools that are mostly finished (70.5% vs. 75.2%). Of the 2026 errors, 42 are selector misses and 42 never had a right answer in the pool. In the selector-miss pools only 17% of candidates are right, so the right answer is a small minority. Diagram questions dominate: 34.8% of them are selector misses, vs. 13.8% of text-only questions.

Selectors (Table XII, Fig. 4). All of these were chosen by LOYO:

- *S2:* the v3 ranker without its length features.
- *S3:* a neural verifier. This is a ModernBERT-base cross-encoder [16], fine-tuned on (English transcript, answer, solution tail; 1,024 tokens) with one fold per training year.
- *S4:* the v3 ranker with the neural score as an extra feature.
- *S5:* S2–S4 retrained with 3,184 new v3-style training candidates (398 training questions, 4 PoT samples each plus their corrections; about 30 GPU-hours). The training pools then match the test-time solutions.

TABLE XII
PHASE 4 SELECTORS, LOYO SELECTION ACCURACY (%) ON 2019–2024 (NO TEST DATA). *v3 pools*: THE 16-ANSWER TRAINING POOLS OF VERIFIER V3; *+W2*: RETRAINED WITH THE NEW V3-STYLE POOLS (S5), EVALUATED ON THE V3 POOLS AND ON THE NEW POOLS. EACH RUN USES ITS BEST OF MAX-PROB / WEIGHTED / LOG-ODDS VOTE ON THE V3 POOLS (METHOD: BASE RUN). GATE (FIXED IN ADVANCE): BEAT THE V3 RANKER ON EVERY POOL KIND.

Selector	v3 pools	+W2: v3	+W2: new	Method	Gate
v3 ranker (refit)	62.0	61.2	59.5	max-prob	ref.
S2 no length	59.1	61.6	60.1	log-odds	fail
S3 neural	57.0	58.7	56.8	weighted	fail
S4 stacked	60.3	61.2	58.8	max-prob	fail
Majority vote	53.7	53.7	55.8		
Oracle	78.5	78.5	71.6		

None passes its gate. The neural verifier does learn a signal: candidate AUC is 0.787 (0.744–0.817 by year), and as a vote weight it beats majority (57.0% vs. 53.7%). But it ranks worse than the handcrafted ranker (62.0%). Dropping the length features costs accuracy (59.1%), consistent with the diagnosis. Matched pools do not help either: the ranker refit with them reaches 61.2%, within the noise of about 1 point we measured for ranker LOYO on these 242 pools. As pre-registered, no Phase-4 selector is scored on test. The one-shot evaluation reproduces the v3 selector (58.8%) and majority (59.3%) on 2026.

Retrieval (Table XIII).

- *Matched seeds.* RAG runs now reuse the no-RAG sampling seeds. A question without exemplars therefore gets

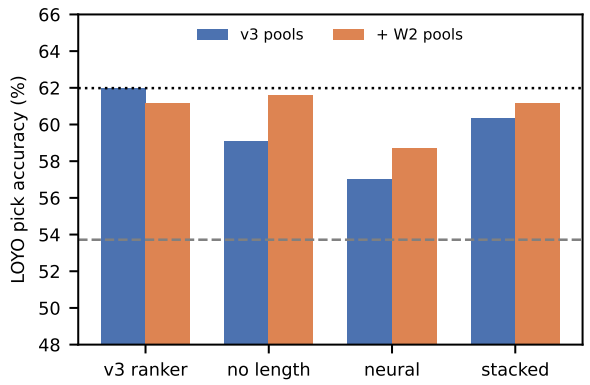


Fig. 4. Phase 4 selectors: LOYO pick accuracy on the 16-answer training pools, trained on the v3 pools or with the new v3-style pools added. Dotted: the v3 ranker; dashed: majority vote.

TABLE XIII
RAG IN PHASE 4, PAIRED DIFFERENCES (POINTS, 95% CLUSTER-BOOTSTRAP CI). TOP: 2024 DEV YEAR WITH MATCHED SEEDS, 204 QUESTIONS, RAG FIRES ON 31. BOTTOM: POST-HOC 2026 POT WITH THE EXTENDED KB (†RIGHT COLUMN: QUESTIONS WHOSE RETRIEVAL CHANGED). LOYO COVERAGE AT PROXY PRECISION 0.70: KB_V4 13.5% VS. KB_V3 15.0%.

Comparison	All	RAG fires
Solutions − no RAG	−0.2 [−1.0, +0.4]	−1.6 [−6.2, +2.5]
Cards − no RAG	+0.3 [−0.2, +0.9]	+2.0 [−1.6, +6.2]
Cards − solutions	+0.6 [−0.1, +1.3]	+3.6 [−0.8, +8.5]
kb_v4 − kb_v3, 2026†	+0.0 [−0.4, +0.4]	+0.0 [−6.2, +7.3]

identical outputs, and the RAG effect is measured only where RAG fires (31 of 204 dev questions).

- *Method cards.* The worked-solution exemplar is replaced by a short card: method, when it applies, plan, key equations and a common mistake, with the answer removed. On the 2024 dev year, where RAG fires, cards beat solution exemplars by +3.6 [−0.8, +8.5] points (P($\Delta \leq 0$)=0.07). The lower bound is below zero, so by the inclusion rule cards are not adopted.
- *Open-licensed KB.* We added 207 verified OlympiadBench [17] solutions (Apache-2.0; text-only, numeric answers). LOYO coverage at proxy precision 0.70 falls from 15.0% to 13.5%: the new near neighbours raise the tuned threshold, and they give only 11 hits themselves. On 2026, retrieval changes for 12 questions, and PoT accuracy is unchanged (+0.0 [−0.4, +0.4]). Coaching material was ruled out for licensing and leakage reasons.
- *Mixed pools.* Half-RAG, half-plain pools give 52.5% majority-vote accuracy on 2026 vs. 52.0% (RAG) and 51.0% (plain).

Takeaway. Each Phase-4 idea addressed a measured weakness and was tested by a rule fixed in advance. None improves on v3. The 2026 selection gap is not explained by length shift. A stronger learned verifier does not close it, and neither do matched training pools. Most remaining errors are perception failures on diagram questions, where the right answer is rare or absent in the pool.

TABLE XIV
WHERE THE V3 GAIN OVER MATCHED SELF-CONSISTENCY COMES FROM (%; DIFFERENCES IN POINTS, 95% CLUSTER-BOOTSTRAP CI). POOL GAIN: MAJORITY VOTE ON THE V3 POOL − MATCHED SC; SELECTION GAIN: V3 PICK − MAJORITY ON THE SAME POOL. FROZEN V3 VERIFIER AND EXISTING POOLS ONLY. MIXED POOLS HOLD BOTH RIGHT AND WRONG ANSWERS (NUMBER OF POOLS IN BRACKETS).

	2025	2026
Matched self-consistency	58.4	52.5
Majority vote on the v3 pool	65.8	59.3
v3 pick (verifier v3)	72.1	58.8
Pool gain	+7.4 [+0.0, +15.3]	+6.9 [+2.0, +11.8]
Selection gain	+6.3 [+1.6, +11.1]	−0.5 [−2.9, +2.5]
Total gain	+13.7 [+5.8, +21.6]	+6.4 [+1.5, +11.3]
Candidate AUC, all	0.872	0.902
Candidate AUC, mixed pools	0.796 (129)	0.748 (121)

VIII. FINAL DIAGNOSTICS (PHASE 4B)

Phase 4 left two questions open. Why does selection fail on 2026, and would a better reading of the figure put more right answers into the pool? We answered them with three small checks. A third pre-registration (`V4B_PREREG.md`), committed before any Phase-4b run, fixed each check and its gate. Phase 4b has its own config, outputs and branch; the Phase-4 outputs are checksummed and read-only. Each check is reported whatever its outcome, and only a check that passes its gate on 2019–2024 would be scored on 2026, once. 2025 and 2026 numbers are post-hoc.

C: where the gain comes from (Table XIV, Fig. 5). We split the gain of v3 over matched self-consistency into a *pool gain* (majority vote on the v3 pool − matched SC) and a *selection gain* (v3 pick − majority vote on the same pool). This uses the frozen verifier and the existing pools only. The pool gain is about the same in both years (+7.4 [+0.0, +15.3] and +6.9 [+2.0, +11.8] points). The selection gain falls from +6.3 [+1.6, +11.1] to −0.5 [−2.9, +2.5]. So the verifier is the part that does not transfer. Its candidate AUC is even higher on 2026 (0.90 vs. 0.87), but that AUC mostly separates easy questions from hard ones. Inside pools that hold both right and wrong answers, the only place where selection matters, the mean AUC falls from 0.80 to 0.75. A high global AUC is therefore no evidence of good selection.

A: an answer-level verifier. The v3 ranker scores candidates one at a time. We instead trained on one row per *distinct answer* in a pool, labelled by whether that answer is right, and picked the most probable answer. The features are fixed in the pre-registration: vote share, vote rank, gap to the top answer, original vs. corrected supporters, the supporters' mean code success, truncation, length and critic flags, and question type (5,335 answer rows from 15,280 training candidates). Logistic regression and four XGBoost settings were compared, and the one with the best *worst* LOYO year was chosen (xgb_d3_t300). It reaches 57.9% (worst year 50.0%), vs. 62.0% (52.5%) for the v3 ranker and 53.7% for majority

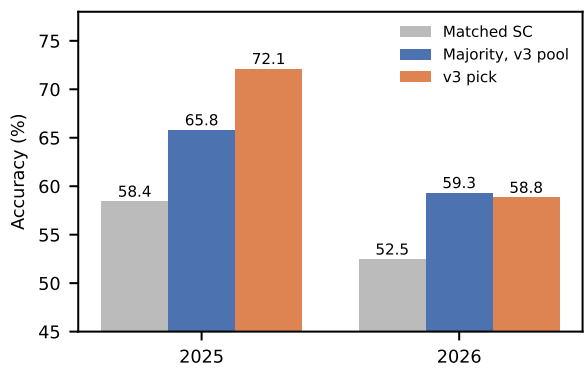


Fig. 5. Gain decomposition (Table XIV). Majority vote on the v3 pool beats matched self-consistency by about 7 points in both years. Picking with verifier v3 adds 6.3 points on 2025 and nothing on 2026.

TABLE XV
DIAGRAM-FIRST TRANSCRIPT ON THE 2024 DEV YEAR (%; DIFFERENCES IN POINTS, 95% CLUSTER-BOOTSTRAP CI). EXISTING: POT WITH THE V3 TRANSCRIPT, SAMPLES 0–7; MIXED: SAMPLES 0–3 OF IT PLUS SAMPLES 4–7 SOLVED FROM THE DIAGRAM-FIRST TRANSCRIPT WITH THE SAME SEEDS; TOP CELLS: EXISTING / MIXED. BOTTOM: SAMPLES 4–7 ONLY, CELLS: CURRENT / DIAGRAM-FIRST READING. GATE (FIXED IN ADVANCE): MAJORITY@8 DIFFERENCE > 0 WITH LOWER BOUND > 0.

	All (204)	Diagram (59)	Text (145)
Majority@8	65.7 / 63.7	42.4 / 39.0	75.2 / 73.8
Diff. (gate)	−2.0 [−5.4, +1.0]	−3.4 [−8.5, +0.0]	−1.4 [−6.1, +2.8]
Oracle@8	79.9 / 78.4	61.0 / 59.3	87.6 / 86.2
Distinct answers	3.05 / 3.18	4.05 / 4.24	2.64 / 2.75
Per sample	55.0 / 51.6	29.7 / 24.6	65.3 / 62.6
Diff.	−3.4 [−6.9, −0.1]	−5.1 [−10.4, −0.4]	−2.8 [−7.0, +1.4]
Truncated	37.3 / 42.6	61.0 / 67.8	27.6 / 32.4
Tokens (k)	3.5 / 3.9	5.3 / 5.7	2.8 / 3.1

vote. It fails the gate and is not scored on test. Aggregating the supporters beats voting, but it loses the detail of single solutions that the ranker's pick relies on.

B: a diagram-first transcript (Table XV). All samples of a question share one transcript, and its figure description is a single line. We added a second, itemised reading of the image (every label, value with unit, arrow, connection and axis) and solved half of each pool from it. Those samples use the same seeds as the reference samples, so only the transcript differs. On the 2024 dev year the mixed pool changes majority@8 by −2.0 [−5.4, +1.0] points, so the gate fails and 2026 was not run. Matched per-sample accuracy falls by −3.4 [−6.9, −0.1] points, most on diagram questions (29.7% → 24.6%). The longer reading leaves less of the fixed token budget for solving: truncation rises from 37.3% to 42.6%. The second reading adds distinct answers but no right ones (oracle@8 79.9% → 78.4%). The diagram gap is therefore not a transcription-format problem. The solver misreads or mis-reasons about the figure whichever description it gets.

Takeaway. On clean data the framework's gain comes from the candidate pool, and that part is stable across years. Selection is fragile: it helps on 2025 and not on 2026, despite a high candidate AUC. Neither a stronger per-candidate verifier (Phase 4) nor answer-level aggregation fixes it, and a more detailed figure description does not help the 8B solver.

IX. DISCUSSION

What holds on clean data. Code-based sampling, cross-model correction and selection beat voting on plain CoT at the same cost, for both v2 and v3. The transcript block and the adaptive budget give a significantly stronger solver.

What does not. Verifier v3 beat majority voting under LOYO and on 2025, but not on 2026. Its features shift with the longer v3 solutions, and its selector gain does not survive the change. Phase 4 shows this is not simply length shift: a fine-tuned neural verifier, a ranker without length features and pools matched to the v3 solutions all stay below the v3 ranker under LOYO. Phase 4b locates the loss: inside pools with both right and wrong answers the verifier's AUC drops from 0.80 to 0.75, while the pool gain holds. The correction gate removes most harmful flips but barely moves accuracy. Retrieval stays a small effect; method cards are a promising but unconfirmed trend.

Remaining failures. Diagram questions and Hindi questions still lag far behind, and about two in five PoT trajectories still end cut off.

X. LIMITATIONS

One substrate model; one clean year of 204 questions (CIs about ±8 points), so differences of a few points are not resolvable. The Hindi 2026 crops are our own. The transcript and the KB texts come from the same VLM. Compute matching counts generated tokens, not prompt tokens or code execution. Phase-4 and Phase-4b numbers on 2025 and 2026 are post-hoc. Phase 4's dev A/Bs have only about 30 questions where RAG fires, and the diagram-first test uses a single dev year.

XI. CONCLUSION

On an exam year the model cannot have seen, a frozen 8B VLM with a code sandbox, cross-model correction and answer selection improves accuracy from 44.3% to 58.8% and beats compute-matched self-consistency by +6.4 points. The v2 framework's claim reproduces on the clean year (+6.4). v3's pre-registered changes make the solver stronger, but its total advantage over v2 is not significant, and its verifier does not beat majority voting there. Phase 4 rules out the obvious fixes on the selection side, and Phase 4b shows that the gain comes from the candidate pool while a more detailed figure description does not help. The next steps are better perception for diagrams and Hindi, which would put the right answer into more pools, and larger dev sets to test retrieval changes such as method cards.

REPRODUCIBILITY

Code, configs and tests are in `mmjee-reasoner/`. The frozen v3 config is `configs/v3_final.yaml` (`V3_FREEZE.md`). Evaluation uses `configs/v3_eval.yaml` and `configs/v3_eval_v2fw.yaml`, and

`scripts/final_report_v3.py` regenerates every v3 table, macro and figure. Phase 4 is isolated in `artifacts_v4/` and `configs/v4*.yaml` (`V4_PREREG.md`, `docs/phases/PHASE4.md`); `scripts/v4_eval.py` is the one-shot test evaluation and `scripts/final_report_v4.py` writes Sec. VII. Phase 4b is isolated in `artifacts_v4b/` and `configs/v4b.yaml` (`V4B_PREREG.md`, `docs/phases/PHASE4B.md`); `scripts/v4b_analysis.py` computes it and `scripts/final_report_v4b.py` writes Sec. VIII.

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